

What was tried, what failed, what changed.

Four pivots from default assumption to measured operator. Each pivot replaced an unverified habit with a measurement.

Exp A · Ch. 6

Autonomous spectral radius → driven Lyapunov exponent

Tried

Cite $\rho(W) < 1$ (the field's default for ESN stability) as evidence the reservoir is in the usable regime.

Failed

$\rho(W)$ is an autonomous, undriven property of W . It says nothing about how the reservoir behaves under real ERP input. $\rho(W) < 1$ does NOT imply the reservoir contracts along driven trajectories.

Changed

Compute the driven Lyapunov exponent λ_1 via the Benettin algorithm along 3,165 real ERP trajectories. Stability is now a MEASURED property of the system in operation, not a cited property of W in isolation.

$\rho(W) = 0.2647549015$ (autonomous); driven λ_1 measured on 3,165 trials.

Exp C · Ch. 6

Hyperparameter chosen a priori → operating-regime measurement

Tried

Set leak rate $\beta = 0.05$ from a biological time-constant argument and use it without measuring the reservoir's memory-capacity response across β .

Failed

Measured MC peak is at $\beta^* \approx 0.012$ ($MC \approx 0.835$), NOT at the dissertation's chosen $\beta = 0.05$ ($MC \approx 0.763 \approx 91\%$ of peak). The 'a priori' choice does not maximize measured capacity.

Changed

Build the Propagation Operating Characteristic — a measured MC curve over β . Re-anchor $\beta = 0.05$ inside the measured plateau ($0.010 \leq \beta \leq 0.118$ all give $MC \geq 0.75$) under a JOINT constraint: $MC \geq 90\%$ of peak AND the time constant $1/\beta$ is matched to within-trial ERP dynamics — $\beta^* = 0.012$ gives $1/\beta \approx 83$ steps, an integration window that washes out stimulus-locked features, $\beta = 0.05$ gives $1/\beta \approx 20$ steps, which does not. β is not chosen by validation accuracy.

Exp D · Ch. 5

Single global channel permutation → per-trial channel permutation

Tried

Test the spatial-feature claim with a SINGLE global channel permutation π applied to every trial — the natural first attempt and what 'channel permutation' meant in early drafts.

Failed

A flat linear classifier compensates for a single π by learning new weights — cross-validated AUC stays unchanged. The 'wrong null' does not actually test what the dissertation claims about spatial structure.

Changed

Generate an independent π_i PER TRIAL. The classifier now sees a (channel_pos, feature_idx) cell that contains data from a DIFFERENT EEG channel on every trial. Per-channel signal is destroyed; only per-trial overall power survives. The STRICTEST possible spatial null.

500 perms × 5 CV folds. Observed AUC sits well outside the null distribution per class.

Exp D · Ch. 5

Fake clinical labels → stimulus-class demonstration with CLI flag

Tried

Demonstrate the channel-permutation methodology on the dissertation's marquee per-disorder labels (SUD / PTSD / GAD / ADHD) as if the labels were already in hand.

Failed

The session-level pickle that ships with the codebase does NOT carry per-subject disorder assignments. Fabricating, approximating, or proxy-labelling them would have invented a clinical effect rather than measured one.

Changed

Refuse to fake labels. Demonstrate the methodology on stimulus class (negative / neutral / positive affect) — explicitly framed as stimulus-class, NOT a clinical-disorder claim. CLI flag (--label-source csv --clinical-csv ...) swaps to disorder labels the instant the CSV is provided.

Red-wired at make_experiment_4_figures.py:389–394; documented in Exp D §1 'The data caveat'. Methodology is the deliverable; the data swap is one flag.

Each pivot replaced an unverified assumption with a measurement. The deck's defensibility lives here.